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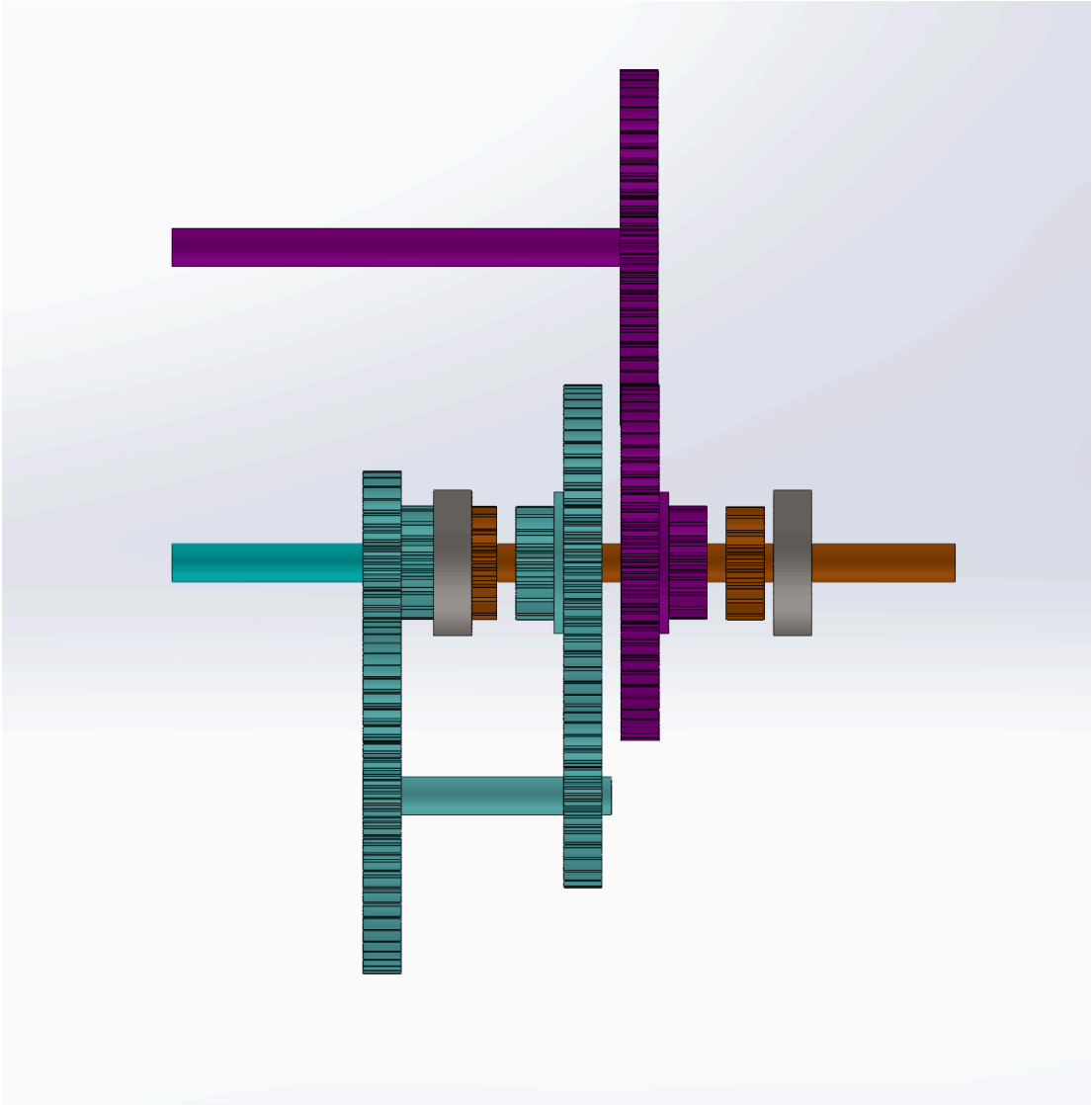
Conner Dial
Alex Shannon
Oscar Barrera
Stanley De Castro
Due: March 25th, 2026

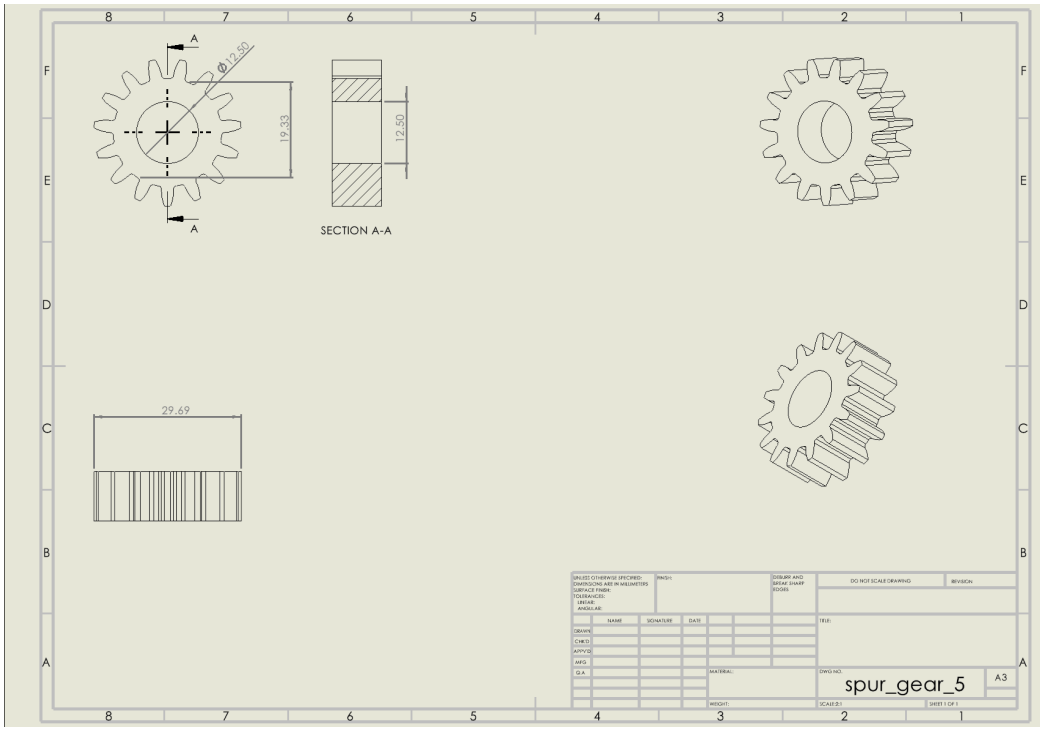
MEE 342: Phase 2 Report

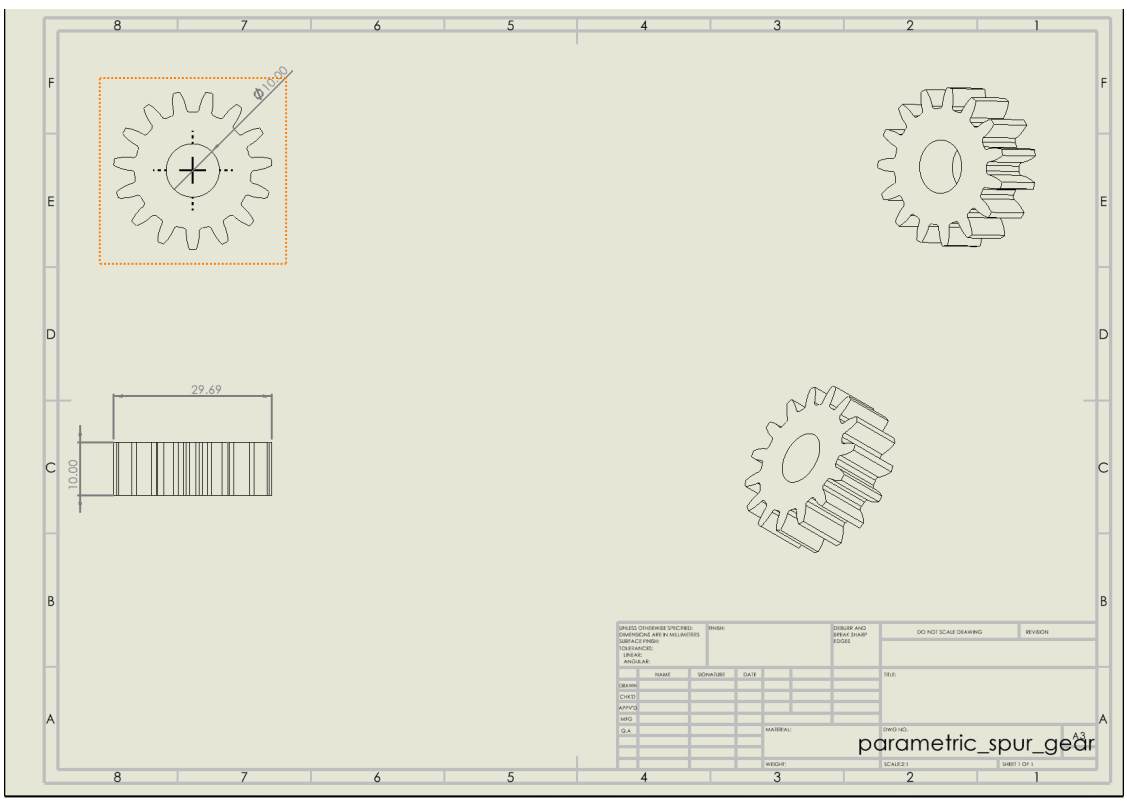
The transfer case is a crucial component in four wheel drive and all wheel drive vehicles that splits power from the transmission and sends it to both the front and rear axles, driveshafts. Its main function is to enable multiple axle power delivery, providing enhanced traction, and allowing drivers to switch between two wheel drive and four wheel drive, or high/low gear ranges.

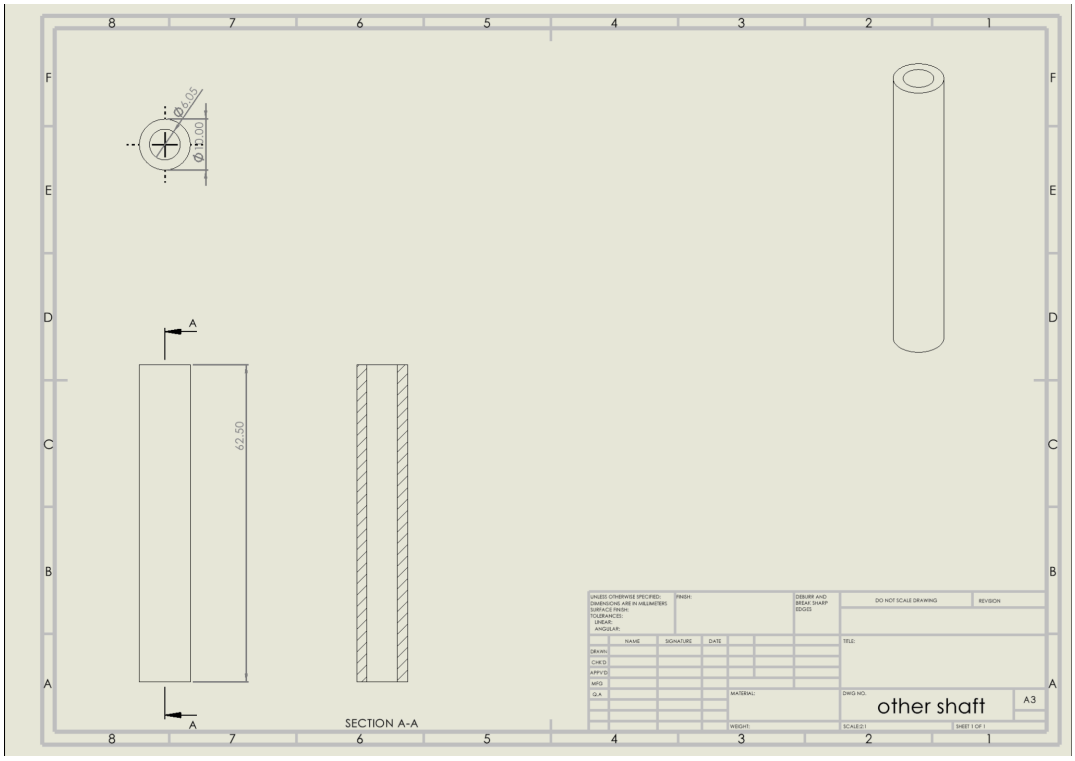
- **Full 3D Assembly**



- **Drawings and Views**



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ANGULAR:
FINISH:
DEBURR AND BREAK SHARP EDGES
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REVISION
NAME
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DATE
DRAWN
CHK'D
APPV'D
MFG
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WEIGHT:
DWG NO.
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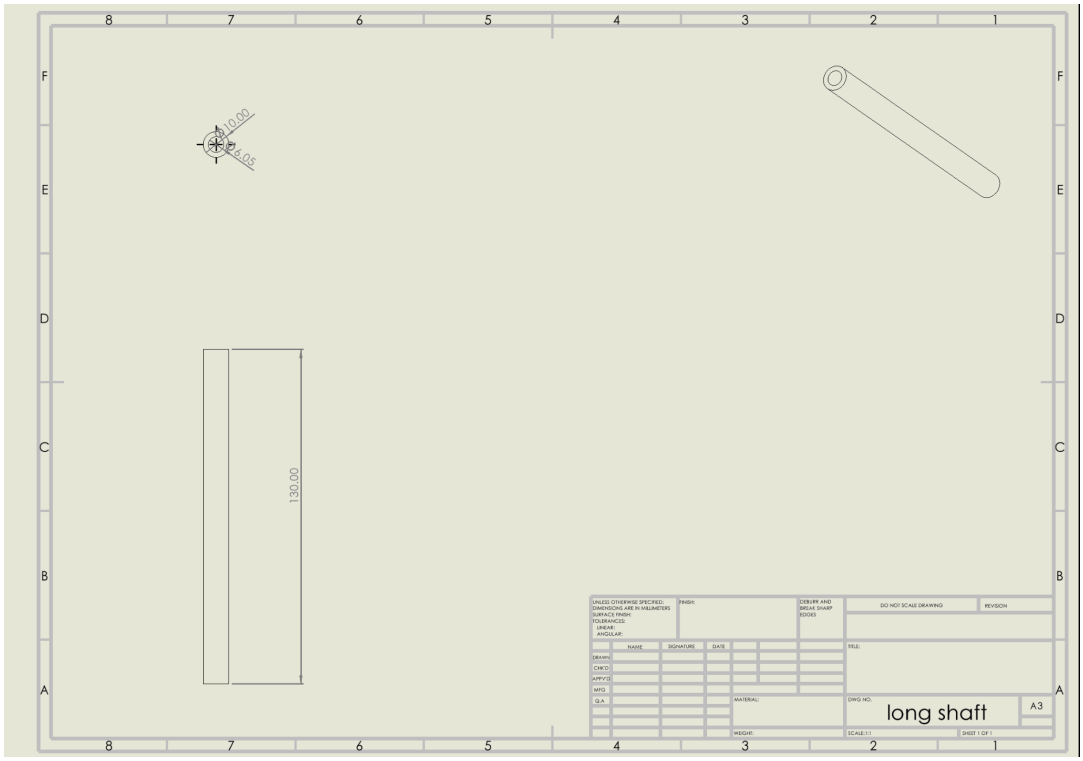
SHEET 1 OF 1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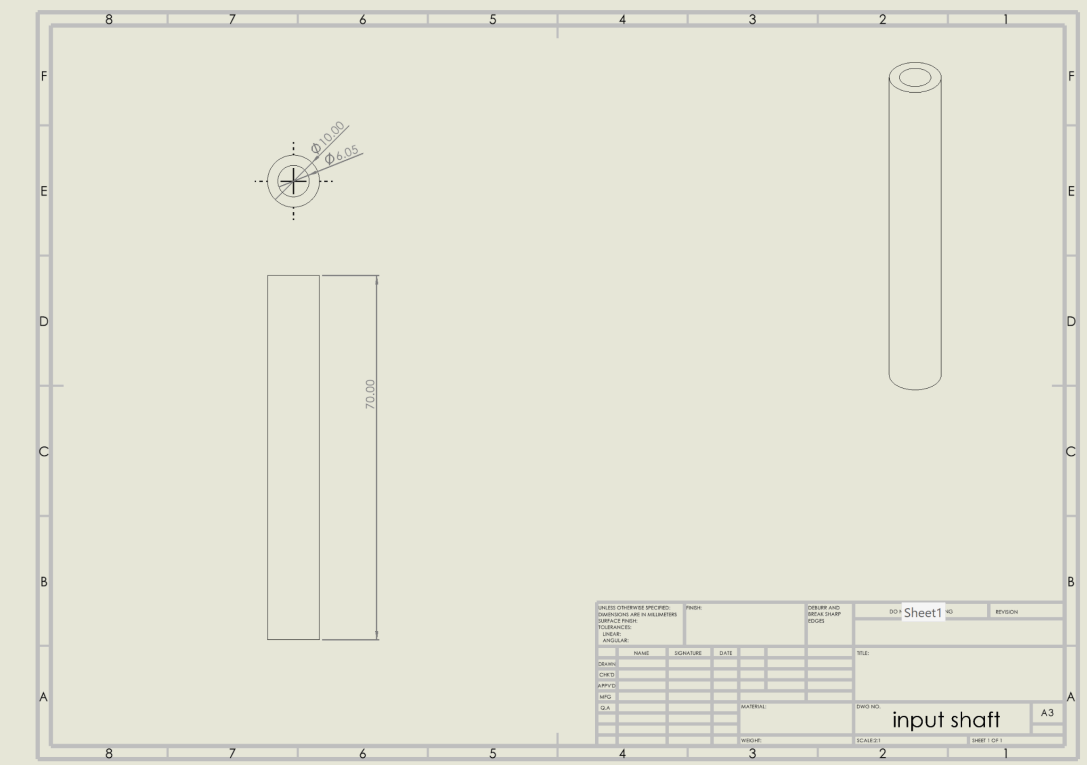
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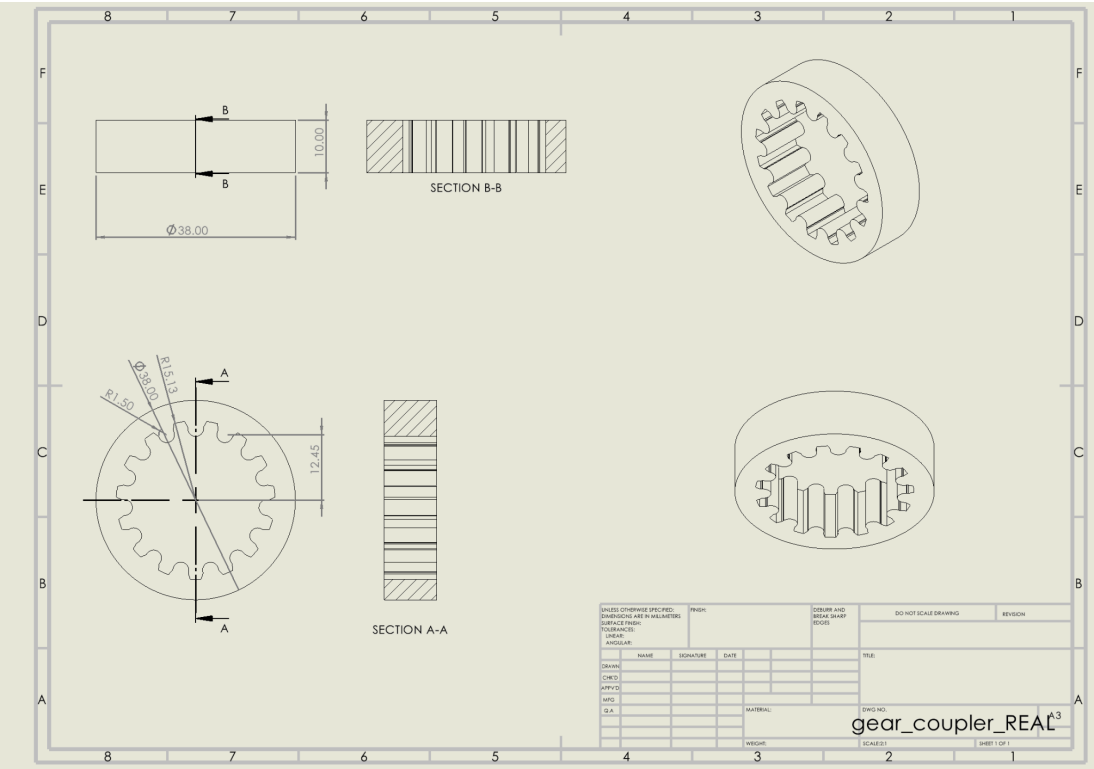
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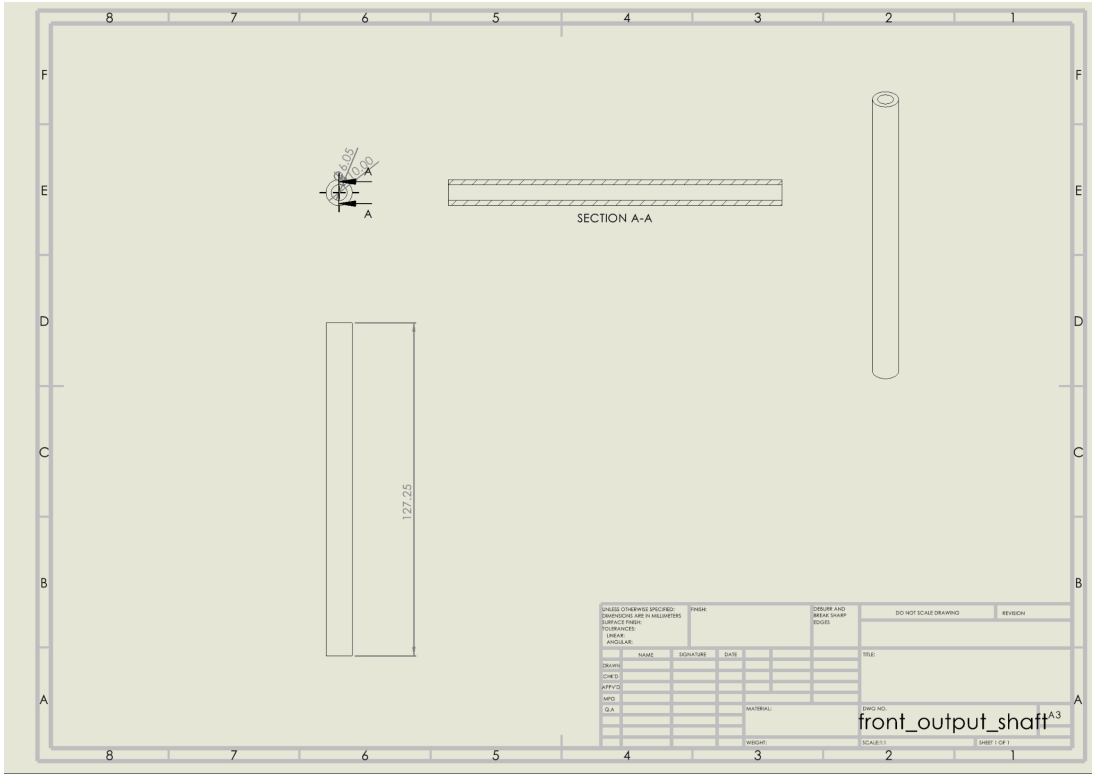
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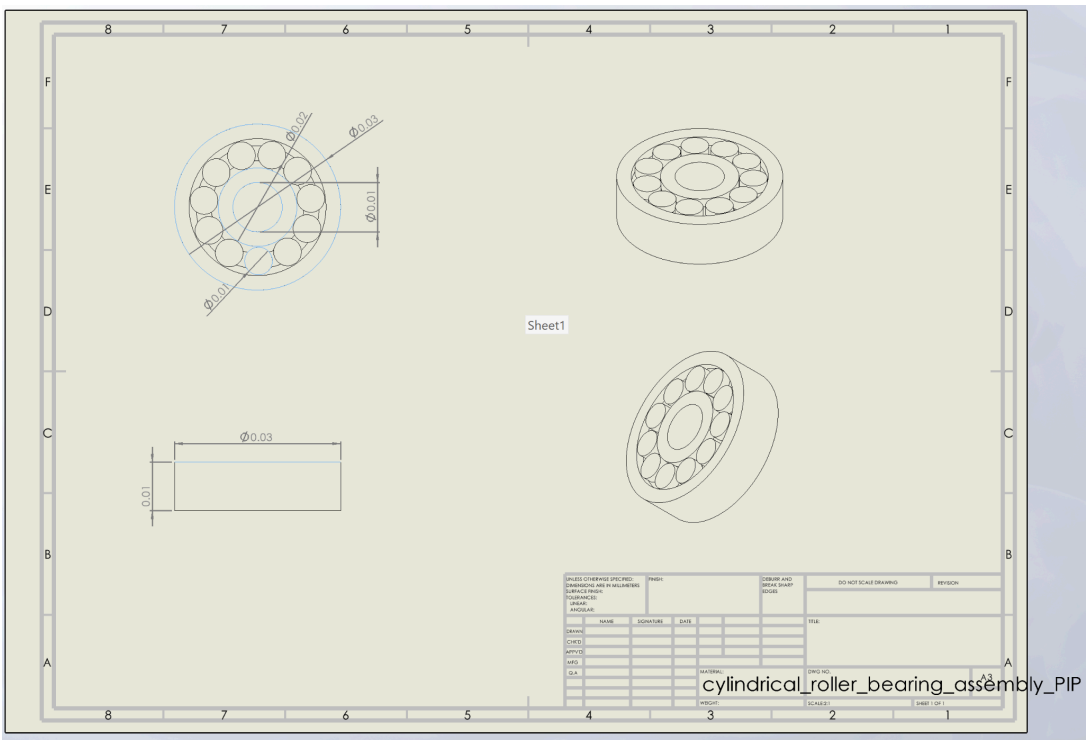
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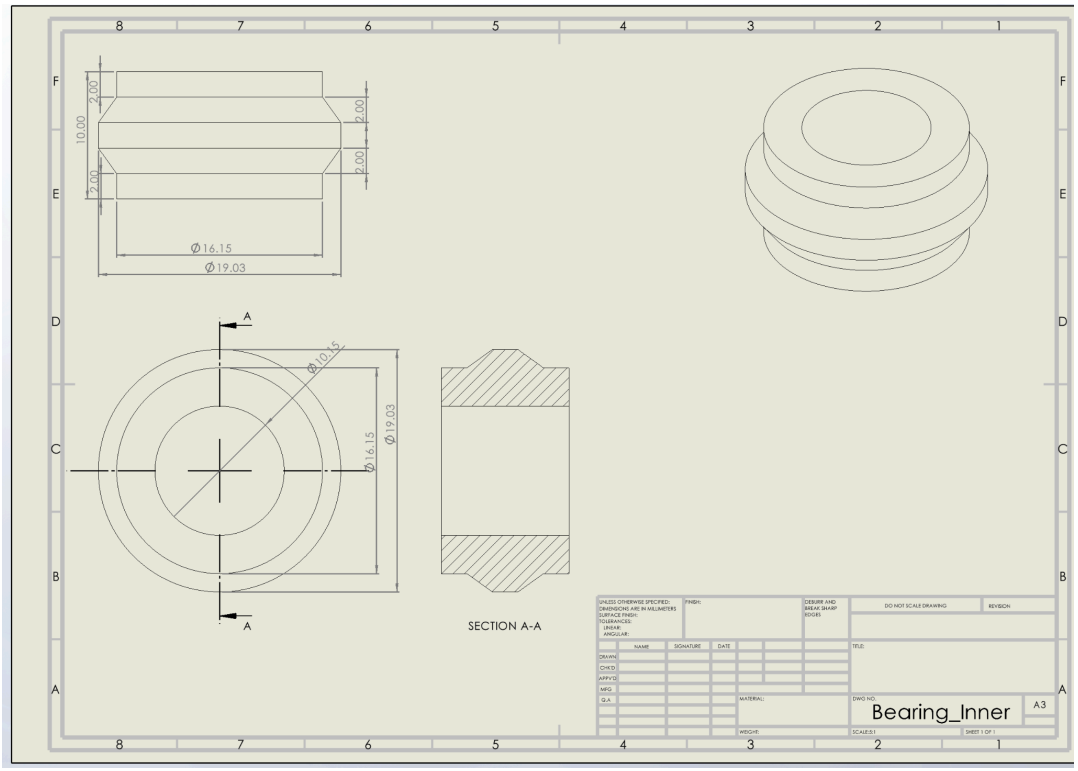
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WEIGHT:
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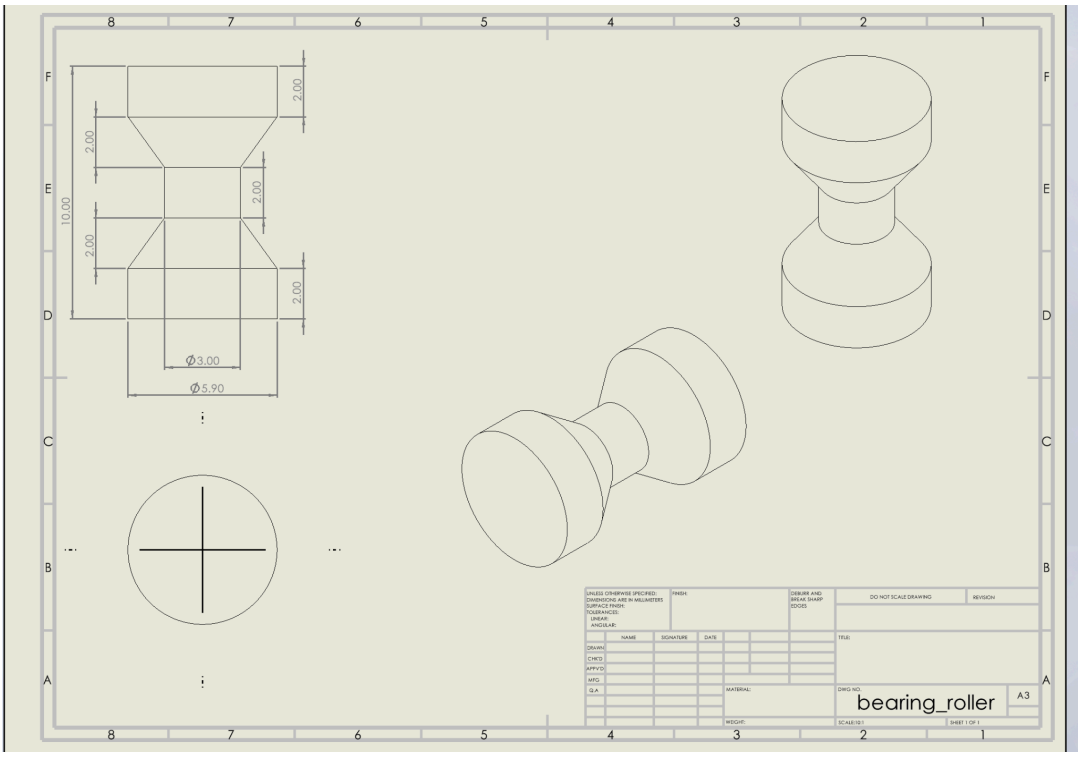
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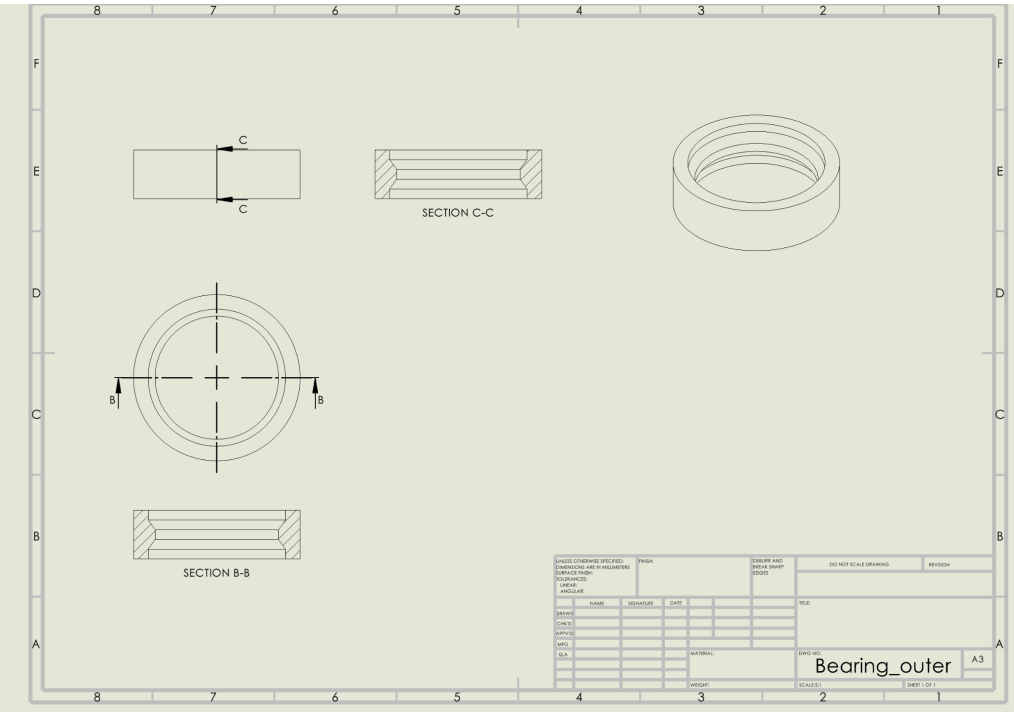
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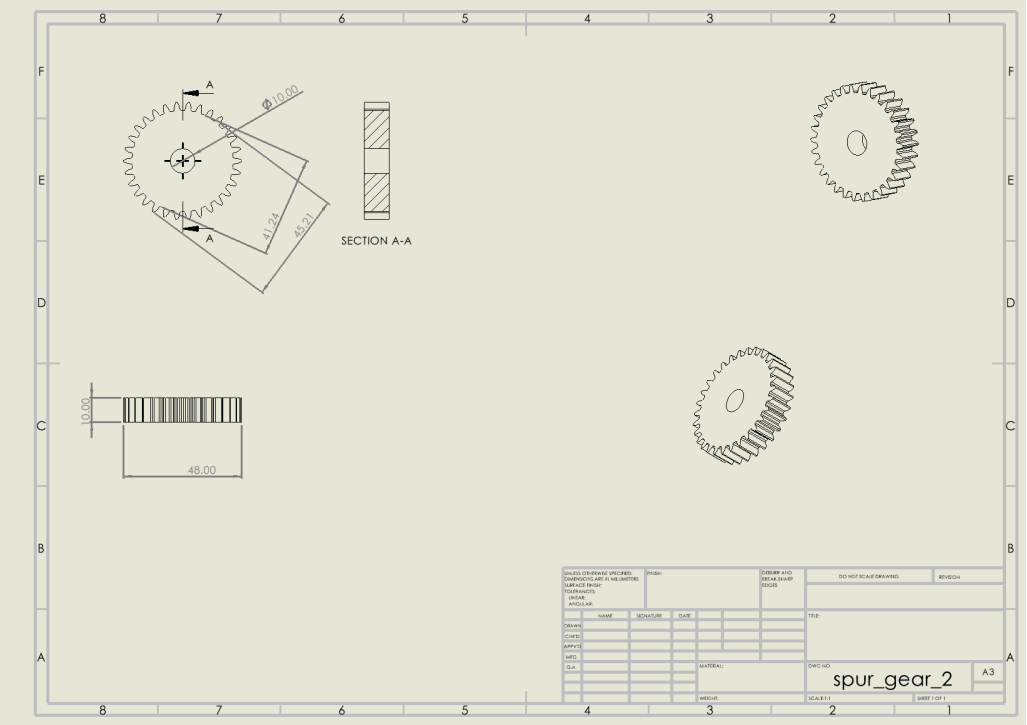
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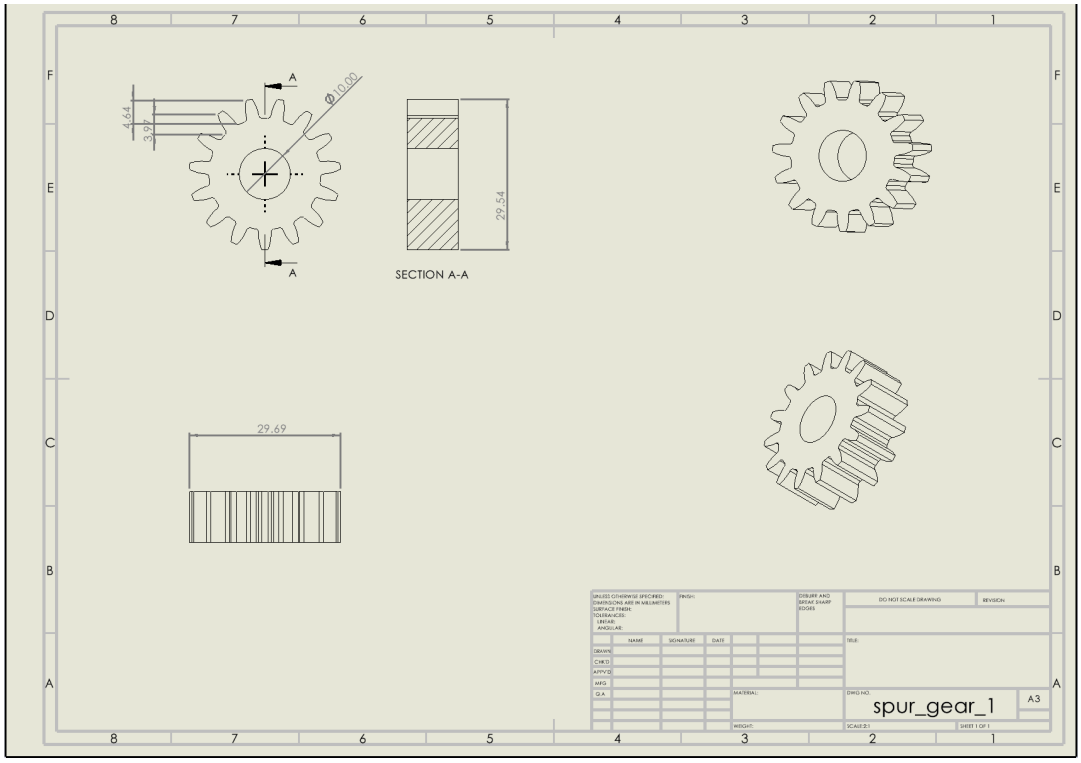
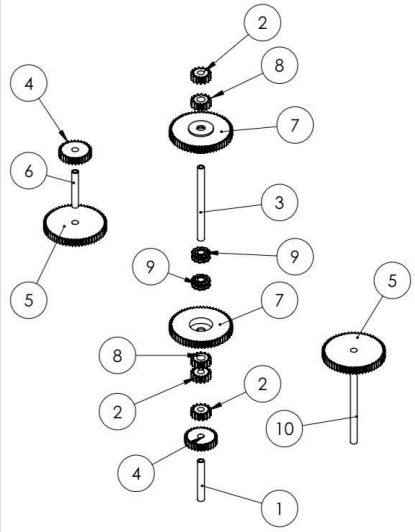
SHEET 1 OF 1





ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Input shaft	4130 Steel	1
2	Spur Gear 1	4140 Steel	3
3	Long Shaft	4130 Steel	1
4	Spur Gear 2	4140 Steel	2
5	Spur Gear 3	4140 Steel	2
6	Medium shaft	4130 Steel	1
7	Spur Gear 4(W/bearing)	4140 Steel	2
8	Spur Gear 5	4140 Steel	2
9	Cylindrical Roller Bearing Assembly	301 Stainless Steel	2
10	Front Output Shaft	4130 Steel	1

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL ±
ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL
FINISH
DO NOT SCALE DRAWING

	NAME	DATE
DRAWN		
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		

COMMENTS:

TITLE:

SIZE A | DWG. NO. Assem1 | REV

SCALE: 1:12 | WEIGHT: | SHEET 1 OF 1

NEXT ASSY | USED ON

APPLICATION

PROPRIETARY AND CONFIDENTIAL

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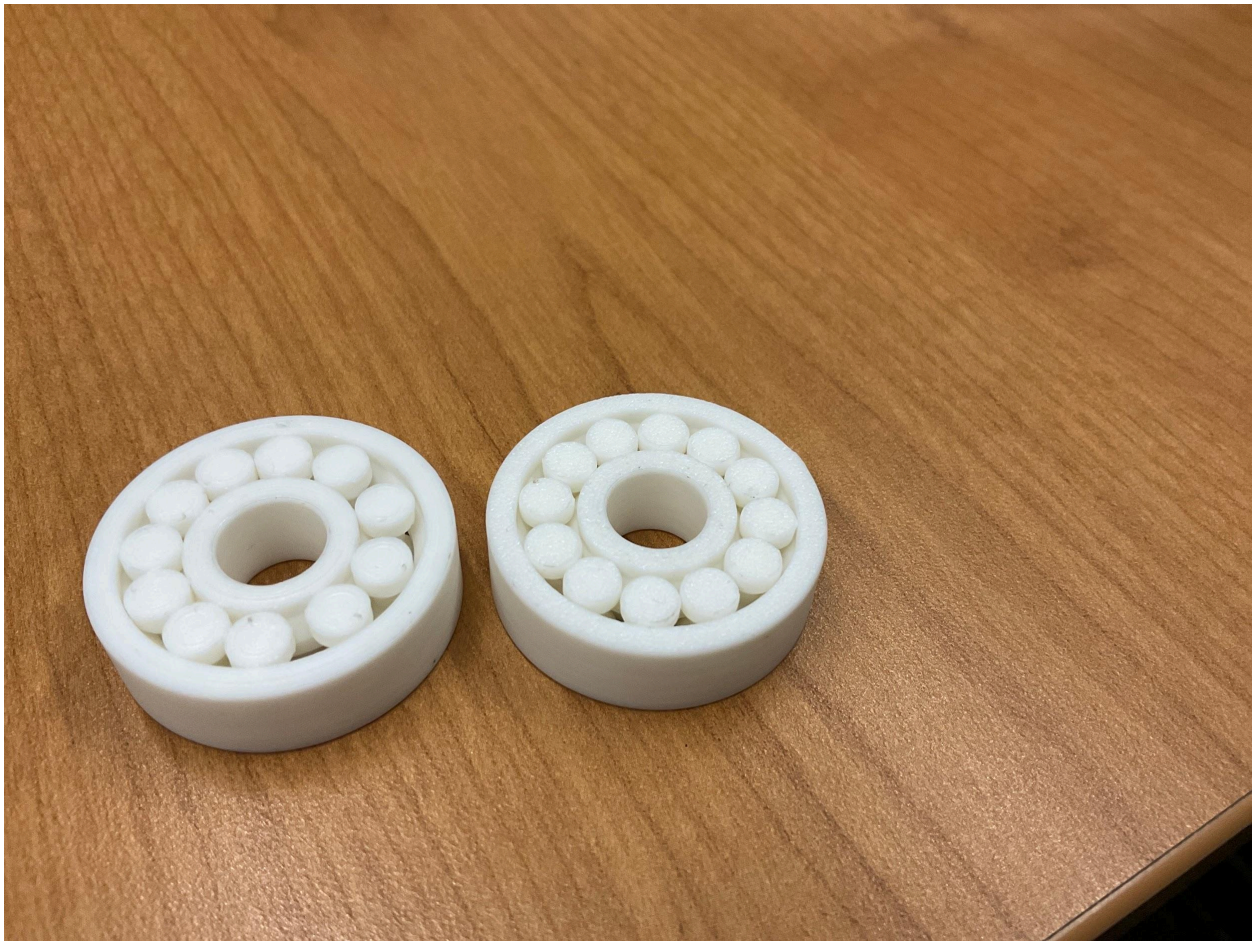
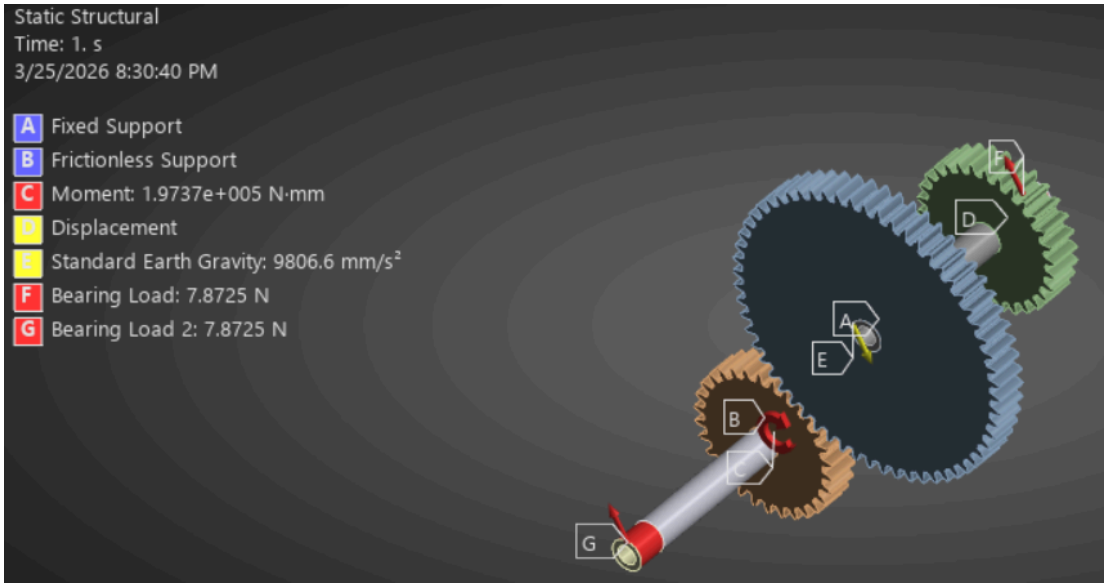
- **Printability**

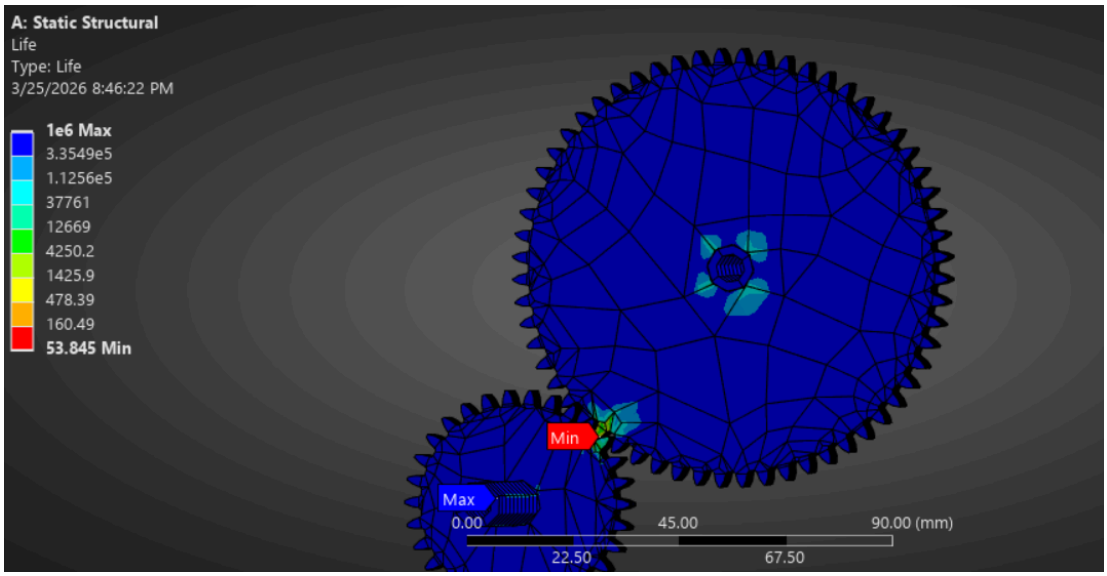


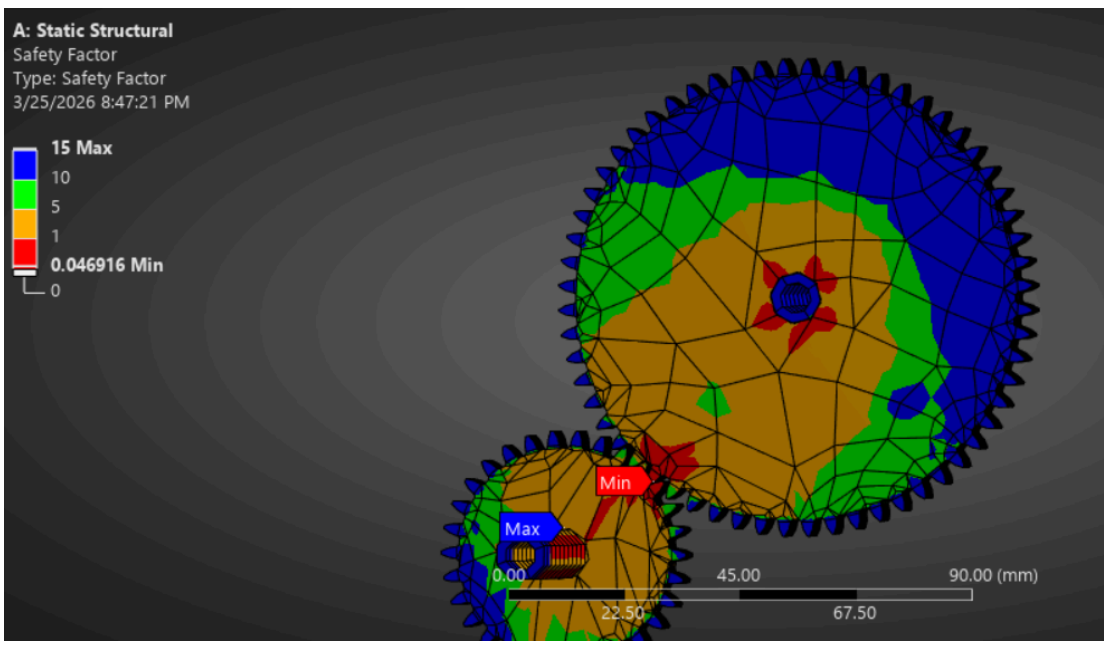
The transfer case was designed with 3D printability in mind by ensuring that all components fit within the available printer build volume and by using practical feature sizes and clearances for the selected printing process. To validate the design, prototype ball bearings were 3D printed and inspected after fabrication. The print bearings were structurally solid and retained their intended geometry, indicating that the design is feasible for physical prototyping. This print test also provided confidence that the chosen tolerances and part dimensions are suitable for assembly and functional evaluation.

- **Ansys Analysis**



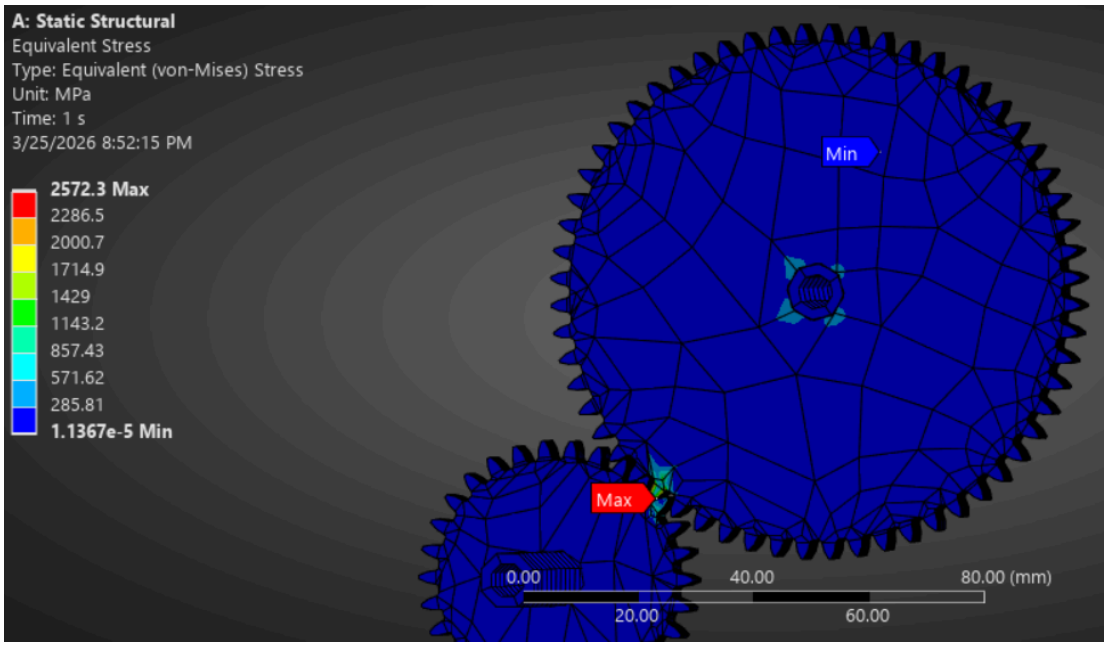
This was the setup we used to conduct the analysis on, the shaft that has the moment applied to it is the input shaft. In this analysis we wanted to find the maximum allowable stresses for bending and contact at the gear tooth, at the time when the system starts moving from rest.

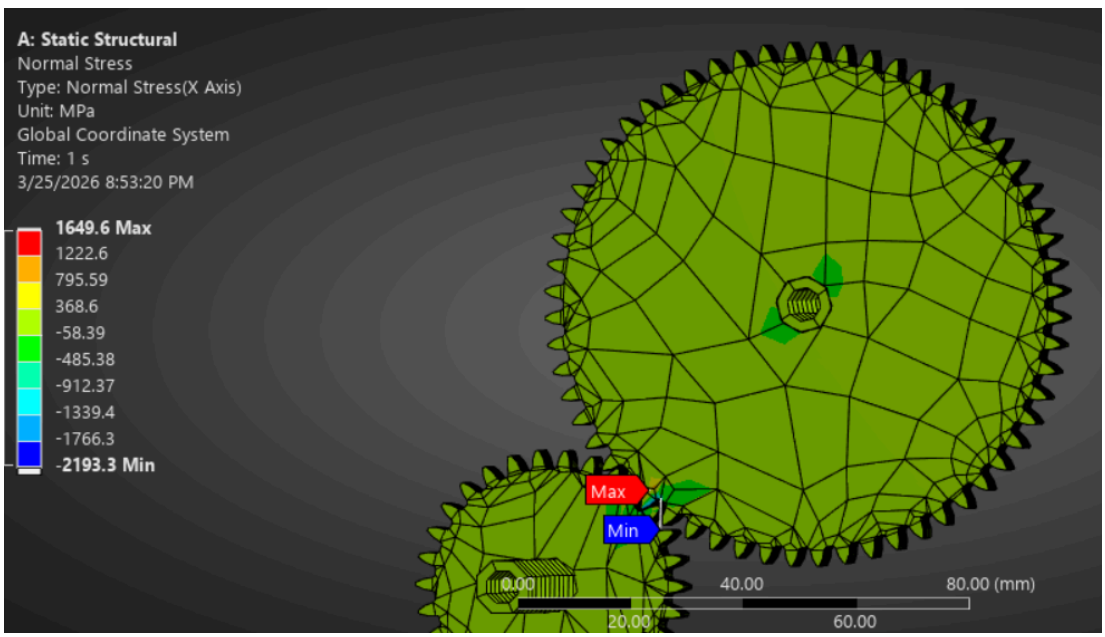




Global Safety Overview

A fatigue analysis was performed on the gear part of the system, while we did expect the teeth to have the lowest lifetime, we did not expect it to be this low, we believe that this is due to the extreme torque we are applying to the system, so we will need to lower that torque applied. Once torque is reduced for a more realistic approach, we can expect a factor of safety of at least 2





The bending and von Mises stresses show that the torque we chose to apply to the system is way too high, and we must choose a different one to apply because our system will fail. Doing a MATLAB analysis of the bending stress of the gears, gave the following results.

```
Actual Stress: 1256.22 MPa
Allowable Stress: 345.15 MPa
CH Factor: 0.9997
STATUS: FAIL (CONTACT)
```

We will need to consider changes to the system, as well as double checking our analyses to ensure that we are proceeding correctly.

- **Statics Analysis**

$\tau = \frac{Tc}{J}$ $\quad J = \frac{\pi}{2}(r_o^2 - r_i^2)$

$E = 200\,GPa$
$G = 80\,GPa$
$S_y = 435\,MPa$

$\tau = \frac{(750\,Nm)(.019)}{J}$ $\quad J = \frac{\pi}{32}(d_o^4 - d_i^4)$

$d_o = .038\,m$
$d_i = .023\,m$

$J = \frac{\pi}{32}(.038^4 - .023^4)$

$J = 1.77 \times 10^{-7}\,m^4$

$\tau_{max} = 80.40\,MPa$ ~ Torsional Shear Stress

$\theta = \frac{TL}{GJ} \rightarrow \frac{(750\,Nm)(.33m)}{(80\,GPa)(1.77\times10^{-7}\,m^4)} = \frac{247.5\,Nm^2}{(80\times10^9\,\frac{N}{m^2})(1.77\times10^{-7}\,m^4)} = \frac{247.5\,Nm^2}{1416\,Nm^2} = .01747\,rad$

$0.01747\,rad = 1°$

$\sigma_{vm} = \sqrt{3}\,\tau_{max}$
$\sigma_{vm} = 139.3\,MPa$ ~ Static yielding Check

$S_y = 435\,MPa$
$n = \frac{S_y}{\sigma_{vm}} = \frac{435}{139.3} = 3.12$ F.S

Belt force of pulley
$T = (F_t - F_s)r$
$F_t - F_s = \frac{T}{r} \rightarrow \Delta F = \frac{750}{r}$

$M = F_{belt,\,resultant}(.09m)$

$F_{belt,\,resultant} = 0.3F_{tight}$

$F_{belt,\,res} = 1.86\frac{T}{r}$